Agent-Based Models of Financial Markets: A Wolfram Perspective*

Jeffrey D. Varner
R.F. Smith School of Chemical and Biomolecular Engineering
Cornell University, Ithaca, NY 14850
jdv27@cornell.edu

Abstract

Financial markets crash without proportionate external shocks, yet most agent-based models either assume global (mean-field) coupling that erases the spatial structure of contagion, or employ ad hoc behavioral rules that resist systematic analysis. We introduce a minimal model in which traders on a two-dimensional grid follow deterministic Wolfram totalistic cellular automaton (CA) rules with three states—Buy, Hold, Sell—and update from their four cardinal neighbors. Boundary agents inject real market sentiment derived from QQQ ETF minute-bar data via a Bollinger Band filter; interior agents are governed by one of two rules, a majority/herding rule and a contrarian/destabilizer rule, whose mixture fraction f is the single control parameter. The model exhibits a sharp first-order phase transition at a critical minority fraction $f_c \approx 0.145$: below this threshold the market is stable, while above it the system abruptly locks into a sell-dominated regime. At the transition, sell cascades nucleate from localized seeds and propagate as spatially coherent wavefronts, a contagion mechanism that is invisible to mean-field treatments and that connects naturally to the tipping-point phenomenology observed in real markets. Because the full rule space ($3^9 = 19{,}683$ rules) is finite and enumerable, the framework enables systematic screening of agent behaviors—a property unavailable in models with continuous parameters.

1 Introduction

Financial markets crash. They crash without proportionate external shocks, without advance warning from traditional risk metrics, and with a statistical regularity—heavy-tailed return distributions, volatility clustering, and transient negative autocorrelation—that is remarkably consistent across instruments, geographic markets, and sampling frequencies [1–3]. A natural question arises: *what is the minimal microscopic mechanism that is sufficient to produce these phenomena?* If the answer involves the collective dynamics of interacting agents rather than exogenous shocks, then the spatial structure of those interactions—who talks to whom, and how far signals travel—matters at least as much as the decision rules themselves [4].

The ingredients for an answer already exist, though they have not been assembled. On one side, a generation of agent-based models (ABMs) has demonstrated that heterogeneous populations of chartists and fundamentalists can reproduce fat tails and clustered volatility [5, 6], that herding on random graphs is sufficient to generate power-law-tailed returns [7], and that minority-game dynamics [8] and adaptive strategy switching [9, 10] produce non-Gaussian fluctuations without any exogenous noise assumption. On the other side, a mature literature on lattice and spin models—Ising models with market coupling [11, 12], opinion dynamics on chains [13], and the three-state speculative model of Kaizoji [14]—has shown that nearest-neighbor interactions on a regular grid

*Repository: https://github.com/varnerlab/Wolfram-markets-abm-paper.git.

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can produce phase transitions between ordered (crash) and disordered (efficient) regimes. Meanwhile, cellular automata (CA) have been applied to financial markets [15–17], but only with ad hoc rule designs that resist systematic comparison. What is missing is a framework that combines *local* spatial dynamics on a regular lattice with a *complete, enumerable* space of behavioral rules and a *direct* connection to real market data.

In this paper, we close that gap. We model agent interactions on a two-dimensional grid governed by Wolfram totalistic CA rules [18], in which each agent occupies one cell, maintains one of three states—Buy, Hold, or Sell—and updates deterministically from the states of its four cardinal neighbors. The totalistic rule framework provides two properties that are unavailable in prior financial ABMs: strictly local information propagation, so that sell cascades must spread from neighbor to neighbor as spatially resolved wavefronts; and a finite, enumerable rule space ($3^9 = 19{,}683$ rules), enabling systematic screening of agent behaviors rather than ad hoc rule design. We study two complementary configurations (Figure 1). In **Model A** (open), boundary agents observe real QQQ ETF minute-bar data via a Bollinger Band filter, injecting exogenous market sentiment that validates the CA against empirical time series. In **Model B** (closed), the boundary agents respond to an endogenous price formed from the grid's own net order imbalance, creating a reflexive feedback loop that produces crashes without any external input. The single control parameter is the minority (contrarian) agent fraction f, and the central result is a sharp first-order phase transition at a critical threshold $f_c \approx 0.145$: below f_c the market is stable; above f_c the system abruptly locks into a sell-dominated regime. The transition is discontinuous, occurring over $\Delta f < 0.01$, consistent with a tipping-point mechanism rather than a gradual crossover [19]. At the transition, sell cascades nucleate from localized seeds and propagate as coherent spatial wavefronts—a contagion mechanism that is invisible to mean-field treatments and that connects naturally to the growing body of work on early-warning signals for critical transitions in complex systems [19, 4].

Section 2 reviews related work on ABMs, lattice models, and CA in finance. Section 3 introduces the CA rule framework and edge-agent decision model. Section 4 describes the simulation procedure for Models A and B. Section 5 presents the phase diagram and grid dynamics, and Section 6 concludes.

2 Related Work

Our work lies at the intersection of three research traditions: agent-based models of financial markets, lattice and spin models from statistical mechanics, and cellular automata theory. We review each strand below and identify where existing approaches stop short of the connection we propose.

2.1 Agent-Based Models and Stylized Facts

The quantitative study of financial market dynamics has been shaped by a tension between tractable theoretical frameworks and the empirical complexity of observed data. Mandelbrot [2] demonstrated that speculative price changes exhibit heavy tails far beyond Gaussian predictions; Fama [20] showed that raw return series display negligible linear autocorrelation, consistent with the efficient markets hypothesis; yet Schwert [21] documented that the *magnitude* of returns exhibits substantial positive autocorrelation decaying slowly over time—large price changes tend to be followed by large changes regardless of sign, a pattern known as *volatility clustering*. Bouchaud and Potters [22] further documented the asymmetric response of volatility to positive versus negative returns (the leverage effect). Cont [1] provides a systematic review confirming these regularities across asset classes, geographic markets, and sampling frequencies. Collectively, leptokurtosis, the absence of return predictability, and volatility clustering define the benchmark against which any model of market dynamics must be evaluated.

Agent-based models offer a bottom-up route to these stylized facts: rather than assuming the statistical structure of returns, they derive it from the simulated interaction of heterogeneous agents. The Santa Fe Artificial Stock Market [23] established the template of adaptive traders interacting through a centralized order book. Lux and Marchesi [5] showed that a two-type fundamentalist-chartist population reproduces heavy tails and volatility clustering as emergent properties, a result that established ABMs as a credible framework for studying market dynamics. Cont and Bouchaud [7] demonstrated that a simple random-graph herding structure alone is sufficient to generate fat-tailed return distributions, highlighting the role of coordination—rather than agent sophistication—in generating extreme events. The minority game of Challet and Zhang [8] introduced the paradigm of

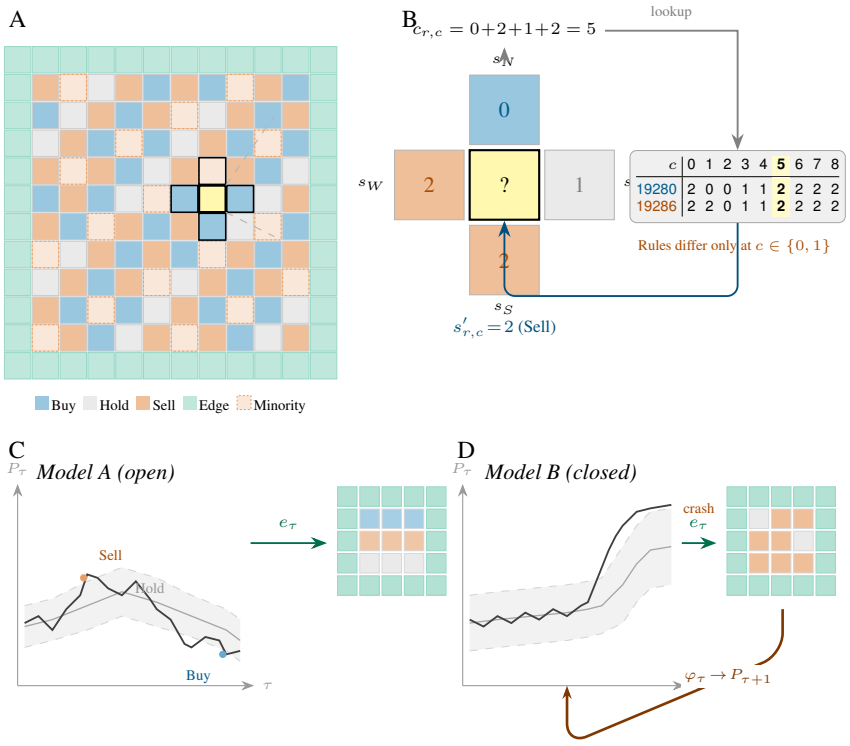


Figure 1: Model architecture. A: Agent grid ($N_r \times N_c$) with three states (Buy, Hold, Sell). Green boundary cells are edge agents; dashed-border interior cells follow the minority rule (19286); solid interior cells follow the majority rule (19280). The highlighted cell and its Von Neumann neighbors are expanded in B. B: CA update: the four neighbor states are summed to produce $c_{r,c}$, which indexes into the rule table to determine the next state $s'_{r,c}$. In this example $c_{r,c} = 5$, so both rules output Sell; the two rules differ only at $c \in \{0, 1\}$. C: Model A (open): exogenous QQQ prices (black trace) fluctuate within a Bollinger Band (gray region). Band breakouts set the edge-agent state, which is injected into the grid boundary. D: Model B (closed): the grid's net order imbalance φ_τ feeds back (orange arrow) to update an endogenous price, closing a reflexive loop.

bounded-rational agents competing via strategic coordination, a framework subsequently shown to produce non-Gaussian fluctuations; Farmer and Joshi [9] demonstrated that standard trading strategies simultaneously deployed by many agents endogenously generate volatility clustering and excess kurtosis without any exogenous noise assumption; and Chen, Tan, and Zheng [24] showed that multi-level herding simultaneously reproduces sector structure and volatility clustering. Harras and Sornette [25] modeled myopic adapting agents whose local herding behavior inflates bubbles that eventually collapse, connecting the ABM literature to crash dynamics.

On the theoretical side, Hommes [10] surveys heterogeneous agent models incorporating adaptive expectations and evolutionary strategy selection; LeBaron [26] provides a systematic treatment of agent-based computational finance; Arthur [27] argues that heterogeneous agent interactions give rise to phenomena—persistent disequilibrium, emergent structure, tipping points—inaccessible to equilibrium theory; and Farmer [28] demonstrates that quantitative ABMs can produce time-series predictions competitive with mainstream macroeconomic models. Battiston et al. [4] make the case that complexity science and ABMs should inform financial regulation. While mechanistically compelling, these models share a persistent limitation: agents interact either globally (mean-field coupling) or on ad hoc network topologies, making it difficult to disentangle the contributions of local spatial contagion from global coordination. The present work addresses this gap by placing agents on a regular lattice with strictly local CA rules, so that every cascade must propagate neighbor-by-neighbor through physical space.

2.2 Lattice Models, Spin Models, and Phase Transitions

The mapping of financial agent interactions onto lattice models draws on the statistical mechanics tradition, where the mathematical machinery for analyzing cooperative phenomena on regular grids—phase transitions, critical exponents, universality classes—is well developed. Ising-type models map agent states (buy/sell) onto spins and impose nearest-neighbor coupling on a lattice [29, 30]; these models naturally capture local herding and can exhibit phase transitions between ordered (bubble/crash) and disordered (efficient) regimes. Bornholdt [11] introduced a two-dimensional Ising model with a global coupling term representing market impact, producing intermittent bubbles and crashes from the interplay of local herding and global frustration across scales. Sznajd-Weron and Weron [13] proposed a one-dimensional opinion model for price formation in which local consensus propagates outward, demonstrating that even the simplest spatial rule can generate non-trivial collective dynamics. Krawiecki et al. [12] showed that attractor-bubbling dynamics on a lattice generate volatility clustering with realistic scaling exponents. Kaizoji [14] extended the Ising framework to three states (buy/hold/sell) to model speculative activity, which is the closest antecedent to our three-state CA formulation.

The crash literature provides the other essential ingredient. Sornette [3] argues that large crashes are the culmination of cooperative herding behavior and can be modeled as critical points analogous to phase transitions in physical systems. The self-organized criticality (SOC) framework of Bak, Tang, and Wiesenfeld [31] suggests that complex systems may naturally evolve toward a critical state where scale-free avalanches occur without parameter tuning; Bak and Paczuski [32] applied this idea to economic systems. Gabaix et al. [33] derive power-law tails from the size distribution of large institutional trades, providing an alternative mechanism for heavy tails that does not require agent heterogeneity. Scheffer et al. [19] propose generic early-warning signals—rising autocorrelation and variance—for systems approaching a tipping point, a framework that has been applied to ecosystems, climate, and financial markets [4]. Farmer et al. [34] demonstrate that even zero-intelligence agents can reproduce aspects of market microstructure when placed in a realistic order-book mechanism, suggesting that the structure of interaction may matter more than the sophistication of individual decision rules.

Our approach generalizes the Ising lattice in two ways: (i) we replace the single coupling constant with Wolfram totalistic CA rules, which provide a richer and fully enumerable space of $3^9 = 19{,}683$ local update functions; and (ii) we couple the lattice to an external or endogenous price signal via boundary agents, enabling direct validation against real market data. The phase transition we observe is discontinuous (first-order) rather than continuous (second-order), and occurs at a sharp critical minority fraction f_c rather than emerging parameter-free as in SOC models.

2.3 Cellular Automata in Complex Systems and Finance

Wolfram [35] classified one-dimensional CA into four behavioral classes—fixed points, periodic orbits, spatial chaos, and complex/undecidable dynamics—and later provided an exhaustive treatment of the subject [18]. Packard and Wolfram [36] extended this classification to two-dimensional totalistic rules on Von Neumann neighborhoods, demonstrating rich pattern-formation dynamics including self-replication, spiral waves, and spatial chaos. The classification program reveals that even deterministic, strictly local rules can produce dynamics ranging from trivially ordered to computationally universal, a property that makes CA attractive as a modeling framework wherever the goal is to understand how microscopic rules generate macroscopic complexity.

The application of CA to financial markets, however, remains sparse relative to the broader ABM literature. Wei et al. [15] proposed a CA model for securities markets in which agent states evolve based on local majority rules and demonstrated qualitative agreement with empirical return distributions. Qiu et al. [16] developed a CA model specifically targeting abnormal fluctuations, showing that local interaction rules can produce heavy-tailed returns. Yang et al. [17] simulated financial markets using CA with heterogeneous update rules and showed that the ratio of contrarian to trend-following agents controls the transition between calm and volatile regimes—a result qualitatively consistent with our phase diagram, though their model uses a different rule framework and does not characterize the transition as a sharp phase boundary. None of these studies uses the full Wolfram totalistic rule framework, and none couples the CA to real market data or resolves the spatial structure of crash cascades.

The present work differs from prior CA-based financial models in three respects. First, we embed agent dynamics in the complete Wolfram totalistic rule space (3^9 rules), enabling systematic enumeration and comparison rather than ad hoc rule design. Second, we couple the CA to real market data (Model A) or an endogenous reflexive price (Model B) via boundary agents, bridging the gap between abstract CA dynamics and empirical financial time series. Third, we identify a sharp first-order phase transition at a critical minority fraction and resolve its spatial structure—nucleation of sell cascades from localized seeds followed by wavefront propagation—connecting the CA dynamics to the tipping-point framework of Scheffer et al. [19].

3 Background

This section establishes the formal ingredients of the model: the CA rule framework that governs interior agent dynamics (Section 3.1), the Bollinger Band filter that translates price signals into edge-agent states (Section 3.2), and the specific rules used in this study (Section 3.3).

3.1 Three-State Totalistic Cellular Automata

We consider a two-dimensional grid of $N = N_r \times N_c$ agents, each in one of three states: Buy ($s = 0$), Hold ($s = 1$), or Sell ($s = 2$). At each discrete time step, every interior agent updates its state deterministically based on the states of its four cardinal (Von Neumann) neighbors. For an agent at position (r, c) with neighbor states $\{s_N, s_S, s_E, s_W\}$, the neighborhood sum is:

$$c_{r,c} = s_N + s_S + s_E + s_W \in \{0, 1, \ldots, 8\}. \tag{1}$$

A *totalistic rule* is a mapping $\mathcal{R} : \{0, 1, \ldots, 8\} \to \{0, 1, 2\}$ that assigns a next state to each of the nine possible neighborhood sums. Since each of the nine outputs can take three values, there are $3^9 = 19{,}683$ distinct rules—each identified by its decimal index:

$$\text{Rule index} = \sum_{j=0}^{8} a_j \cdot 3^j, \tag{2}$$

where $a_j = \mathcal{R}(j)$ is the next state when the neighborhood sum equals j. The key structural observation is that this rule space is finite, enumerable, and complete: every possible deterministic, translation-invariant, totalistic update function on the Von Neumann neighborhood is represented exactly once.

3.2 Edge Agent Decision Model

Boundary cells serve as *edge agents* that inject external information into the grid (Figure 1A, green cells). A fraction f_e of boundary cells, selected uniformly at random with a fixed seed, are designated as edge agents; the remaining boundary cells follow the same CA rule as interior agents.

Each edge agent classifies the current price P_τ relative to an exponential moving average (EMA) and its associated volatility band:

$$\hat{S}_\tau = \hat{S}_{\tau-1} + \alpha\big(P_\tau - \hat{S}_{\tau-1}\big), \tag{3}$$

$$\hat{\sigma}_\tau = \hat{\sigma}_{\tau-1} + \alpha\big(|P_\tau - \hat{S}_\tau| - \hat{\sigma}_{\tau-1}\big), \tag{4}$$

where $\alpha = 1/60$ corresponds to a 60-minute smoothing window. The decision rule is a three-way classifier:

$$e_\tau = \begin{cases} 0 \text{ (Buy)} & \text{if } P_\tau < \hat{S}_\tau - k\,\hat{\sigma}_\tau, \\ 1 \text{ (Hold)} & \text{if } \hat{S}_\tau - k\,\hat{\sigma}_\tau \le P_\tau \le \hat{S}_\tau + k\,\hat{\sigma}_\tau, \\ 2 \text{ (Sell)} & \text{if } P_\tau > \hat{S}_\tau + k\,\hat{\sigma}_\tau, \end{cases} \tag{5}$$

where k controls the width of the hold band. When the price is within k standard deviations of its moving average, the edge agent holds; when the price breaks out above (below) the band, it sells (buys). This is a standard mean-reversion signal widely used in technical analysis, and its simplicity is deliberate: the edge agents provide a minimal, interpretable channel through which real market information enters the grid.

Table 1: Wolfram totalistic CA rule taxonomy. Base-3 encoding reads right-to-left: digit a_i gives the next state when the neighbourhood sum index equals i.

Rule	Base-3 encoding	Interpretation	Role
6158	$(022110002)_3$	Imitation / herding	Homogeneous model; stable waves
19280	$(222110002)_3$	Majority rule	80% crash model; herding
19286	$(222110022)_3$	Contrarian	20% crash model; destabiliser

3.3 Rule Taxonomy

Table 1 lists the three rules used in this study. Rule 19280 is a majority/herding rule: agents tend to adopt the dominant state in their neighborhood, promoting coordination and spatial coherence. Rule 19286 is a contrarian rule that differs from 19280 in a single respect—its response to extreme sell neighborhoods ($c \in \{7, 8\}$). When all four neighbors are selling, a Rule 19280 agent *buys* (damping the cascade), whereas a Rule 19286 agent *sells* (amplifying it). This single-digit difference in the base-3 encoding is the minimal perturbation that converts a stabilizing agent into a destabilizing one, and it is this minimal difference that drives the phase transition studied below.

4 Methods

The background section established the CA rule framework, the edge-agent decision model, and the specific rules under study. We now describe the two simulation configurations and the phase-sweep protocol used to map the crash transition.

4.1 Model A: Open (Boundary-Driven) Simulation

In Model A (Figure 1B), edge agents observe real market data—specifically, one-minute OHLC bars for the QQQ ETF (Invesco QQQ Trust, 2024 calendar year, $T = 23{,}790$ time steps). At each step τ three operations occur in sequence: (i) the typical price $P_\tau = (H_\tau + L_\tau + C_\tau)/3$ is computed from the OHLC bar; (ii) each edge agent evaluates Eq. (5) using the online EMA and volatility from Eqs. (3)–(4); and (iii) all interior agents update simultaneously via their assigned CA rule. The grid is initialized to all-Hold ($s = 1$), so that any Buy or Sell activity must originate from the boundary and propagate inward.

For the heterogeneous (crash) model, a fraction f of interior agents are assigned Rule 19286 (contrarian) uniformly at random, while the remaining $1 - f$ agents use Rule 19280 (majority). The minority agent positions are fixed for the duration of the simulation (quenched disorder), so that the only control parameter is the mixture fraction f.

Simulation parameters. The default grid is 256×256 ($N = 65{,}536$ agents). Edge agents comprise 25% of boundary cells ($f_e = 0.25$, yielding $\sim$512 active edge agents), selected with a fixed random seed for reproducibility. The EMA smoothing parameter is $\alpha = 1/60$ (a 60-minute window) and the band width is $k = 1.0$ standard deviations.

4.2 Model B: Closed (Endogenous Price) Simulation

In Model B (Figure 1C), there is no external price feed; instead, the price is generated endogenously from the grid's own collective state. Let $N_{\text{buy},\tau}$ and $N_{\text{sell},\tau}$ denote the number of interior agents in the Buy and Sell states at step τ, respectively. The net order imbalance is:

$$\varphi_\tau = \frac{N_{\text{buy},\tau} - N_{\text{sell},\tau}}{N_{\text{interior}}}, \tag{6}$$

and the price updates as:

$$P_{\tau+1} = P_\tau \cdot \exp\big(\lambda\, \varphi_\tau\big), \tag{7}$$

where $\lambda > 0$ is the price impact parameter controlling how strongly the aggregate order flow moves the price. Edge agents then evaluate Eq. (5) using the endogenous price $P_{\tau+1}$ with online EMA updates, closing the feedback loop. The result is a fully self-contained system: the grid generates the price, the price drives the edge agents, and the edge agents inject signals back into the grid.

4.3 Phase Sweep Protocol

To map the crash transition, we sweep the minority fraction f from 0.0 to 0.50 in two passes. A coarse sweep with $\Delta f = 0.05$ covers the full range; a fine-grained sweep with $\Delta f = 0.01$ resolves the transition region $f \in [0.08, 0.22]$. For each value of f, the full T-step price series drives a 128×128 grid (chosen for computational efficiency; the transition location is robust to grid size), and we record the sell fraction at every time step. Three summary statistics characterize each run: mean sell fraction, peak sell fraction, and crash frequency (the fraction of time steps at which the sell fraction exceeds 0.50).

5 Results and Discussion

5.1 Grid Dynamics: Stable vs. Crash Regimes

We simulated the 256×256 agent grid under Model A for three values of the minority fraction—$f = 0$ (homogeneous), $f = 0.15$ (near-critical), and $f = 0.20$ (supercritical)—and recorded grid snapshots at four logarithmically spaced time points to capture both the early transient and the late-time steady state (Figure 2).

In the homogeneous model ($f = 0$, top row), the grid reached a dynamic steady state dominated by Buy (blue) and Sell (orange) agents in a fine-grained spatial mixture; the pattern fluctuated locally but never exhibited large-scale coordination, and the sell fraction remained near its equilibrium value of $\sim$0.26 throughout the simulation. At $f = 0.15$ (middle row)—just above the critical threshold—the system initially resembled the homogeneous case, but at $t = 8{,}236$ we observed a sell cascade that nucleated from a localized seed and propagated outward as a spatially coherent wavefront. The orange region grew concentrically until it engulfed the grid by $t = 23{,}790$, which suggests a contagion mechanism qualitatively different from the instantaneous global crashes produced by mean-field models: the crash had a *spatial history*, starting at one location, spreading through a physical medium, and arriving at different parts of the grid at different times (Figure 2, middle row). At $f = 0.20$ (bottom row), the cascade occurred earlier and more rapidly; the grid was already predominantly sell-dominated by $t = 2{,}851$, consistent with a system deep in the supercritical regime.

5.2 Phase Diagram

We swept the minority fraction f from 0.0 to 0.50 and computed three summary statistics—mean sell fraction, peak sell fraction, and crash frequency—for each run (Figure 3). All three order parameters exhibited a sharp discontinuous transition at a critical threshold $f_c \approx 0.145$ (dashed vertical line).

Below f_c, the system remained stable: the mean sell fraction hovered near 0.26 (close to the $1/3$ expected for a uniform random mixture of three states), the peak sell fraction never exceeded 0.30, and no crashes occurred. Above f_c, the system abruptly entered a sell-dominated regime: the mean sell fraction jumped to $\sim$0.50 at $f = 0.15$ and saturated near 0.91 by $f = 0.20$; crash frequency exceeded 95% (Figure 3C). The transition width was less than $\Delta f = 0.01$, and the jump in the order parameter was discontinuous, both hallmarks of a first-order phase transition rather than the smooth crossover that would characterize a second-order (Ising-type) critical point.

These results suggest a clear economic interpretation: a small minority of contrarian agents was harmless, because Rule 19280 (majority) agents damped local sell signals through herding toward the dominant state. But once the contrarian fraction exceeded f_c, Rule 19286 agents were numerous enough to nucleate sell clusters that the majority-rule agents could no longer absorb; instead, the herding mechanism *amplified* the cascade, because the local majority had tipped from buy to sell. The result was a coordination failure—a flash crash triggered by the interaction between two deterministic rules, without any stochastic component, learning, or adaptive strategy switching. The crash was an emergent property of the rule mixture, not of any individual rule.

6 Conclusion

We presented a minimal agent-based model of financial markets in which traders on a two-dimensional grid follow deterministic Wolfram totalistic cellular automaton rules. The model required only two ingredients—a majority/herding rule (19280) and a contrarian/destabilizer rule (19286)—and a single

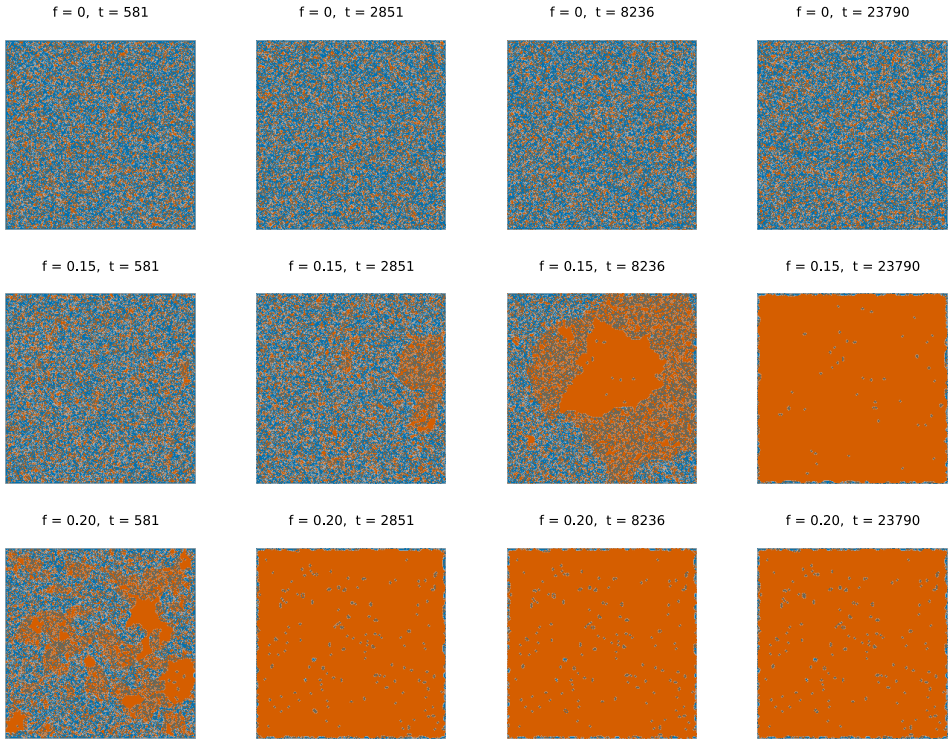


Figure 2: Spatiotemporal dynamics of the 256×256 agent grid under Model A (Rules 19280/19286, QQQ 2024 data). Columns share logarithmically spaced time points; rows vary the minority fraction f. Blue = Buy, gray = Hold, orange = Sell. At $f = 0.15$ (middle row, $t = 8{,}236$), a sell cascade nucleates from a localized seed and propagates as a coherent wavefront before engulfing the grid.

control parameter, the minority agent fraction f. We observed a sharp first-order phase transition at $f_c \approx 0.145$: below this threshold the market remained stable; above it, the system abruptly locked into a sell-dominated regime (Figure 3).

Three features distinguished this framework from prior ABMs and lattice models. First, the CA enforced strictly local information propagation, which produced spatially resolved sell cascades that nucleated from localized seeds and spread as coherent wavefronts (Figure 2). This mechanism—a crash with a spatial history, arriving at different parts of the grid at different times—was absent in mean-field models and highlighted the role of spatial topology in crash formation. Second, the

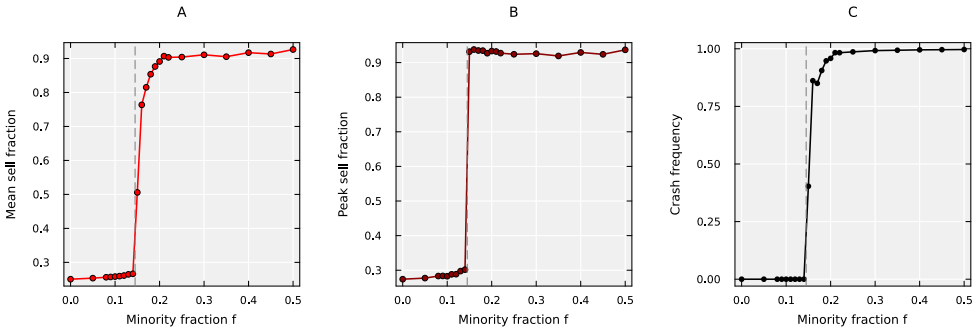


Figure 3: Phase diagram for Model A (Rules 19280/19286, 128×128 grid). **A:** Mean sell fraction vs. minority fraction f. **B:** Peak sell fraction. **C:** Crash frequency (fraction of time steps with sell fraction > 0.50). Dashed vertical line marks $f_c \approx 0.145$. The transition is discontinuous, consistent with a first-order phase transition.

transition was discontinuous, occurring over $\Delta f < 0.01$, which places it squarely within the tipping-point framework of Scheffer et al. [19] and suggests that early-warning signals (rising autocorrelation and variance) may be detectable in the pre-transition regime. Third, the full rule space is finite and enumerable ($3^9 = 19{,}683$ rules), enabling systematic screening of agent behaviors—a property unavailable in models with continuous behavioral parameters.

Limitations. The present study uses a single price series (QQQ, 2024 calendar year) for Model A validation; robustness across instruments and market regimes remains to be established. The CA rules are deterministic and do not incorporate stochastic noise, learning, or adaptive strategy switching—features that are present in real markets and that would enrich the model at the cost of breaking the enumerability of the rule space. The economic interpretation of individual rules (e.g., why a specific digit in the base-3 encoding corresponds to contrarian behavior) remains heuristic rather than derived from utility maximization or observed trader behavior.

Future directions. Model B (closed-loop endogenous price) remains to be fully characterized. Its four-dimensional parameter space (k, w, λ, f) admits an analytical stability analysis via mean-field linearization that may yield a closed-form expression for the phase boundary. A key open question is whether Model B self-organizes to criticality without fine-tuning of f—i.e., whether the reflexive feedback loop drives the system toward its own tipping point, a phenomenon that would connect this framework to the broader literature on self-organized criticality in financial markets [3, 31, 33].

Acknowledgments

TODO: Acknowledgments here.

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A Supplementary Material

TODO: Write appendix content here.